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## Data Governance and Unity Catalog

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### Overview of Data Governance and Unity Catalog

Data Governance is simply the set of rules, roles, and processes an organization uses to keep its data secure, accurate, and private.

In Azure Databricks, **Unity Catalog** is the tool that handles all of this. Instead of setting up security rules inside individual workspaces or configuring complex Azure permissions for every single folder, Unity Catalog gives you one central dashboard to manage security for your entire data ecosystem.

It Provides:

- **Central Control:** You can grant or block access to data using simple SQL commands (GRANT or REVOKE).
- **Lineage Tracking:** It automatically creates a visual map showing where data comes from and how it changes over time.
- **Audit Trails:** It keeps an exact log of who accessed what data and when, which is perfect for security audits.

### Create Unity Catalog Metastore and enable a Databricks workspace with Unity Catalog

To get Unity Catalog running, you need a central warehouse to store all your data definitions. This is called a **Metastore**. It stores metadata (information about data), such as:

- Catalog names
- Schema names
- Table definitions
- Permissions

Steps to Create it in Azure

**Set Up Azure Store** - Unity Catalog needs a place in Azure to save its data. You create an Azure Data Lake Storage Gen2 (ADLS Gen2) container. You also create an Azure Access Connector (a managed identity) so Databricks has a secure way to read and write to that storage container without needing manual passwords.

**Create the Metastore** - Next, you log into the Databricks Account Console (the master admin page). Go to the Catalog tab, click Create Metastore, and give it a name and a region. You then paste the link to your Azure storage container and the Access Connector ID to tie them together.

**Enable Your Workspace** - Finally, while still in the Account Console, you open your new metastore, click Assign to Workspaces, and check the box next to your Databricks workspace. The moment you hit save, Unity Catalog is officially activated for that workspace.

## Overview of 3 level namespace and creating Unity Catalog Objects

Once Unity Catalog is turned on, it organizes all your data using a strict three-tier hierarchy called a **3-level namespace**. When you want to query data, you must type out its full path like this:

***Catalog.Schema.Table***

**Catalog (Level 1):** The highest organizational folder. Companies usually name these after environments (like dev or prod) or departments (like finance or marketing).

**Schema (Level 2):** Also known as a database. This lives inside a catalog and groups related topics together (like sales\_records or customer\_profiles).

**Object (Level 3):** This is the actual asset containing your data. It can be a structured Table, a View, or a Volume (a folder used to store unstructured files like raw PDFs, images, or CSVs).